

TRANSFORMATION

- In all computer graphics system ,the fundamentally ability is to simulate the manipulation of an object in space.
- A computer vision system should allow the programmer to define picture that include a variety of transformation.
- It is a mathematical way of moving or modifying objects without losing their basic structure.
- It is also able to change scale of a symbol.
- But the object is still recognizable after the change.

In computer graphics ,there are two types of transformation:-

1.2D Transformation

2.3D Transformation

The operations which are performed by 2D transformation are as

1. Translation
2. Rotation
3. Scaling
4. Reflection
5. Shear

Translation

A translation is applied to an object by re-positioning it along a straight-line path from one coordinate location to another. We translate a two-dimensional point by adding translation distances, t_x and t_y to the original coordinate position (x, y) to move the point to a new position (x', y') . Where t_x & t_y are translation factors.

$$(x, y) \rightarrow (x + a, y + b)$$

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

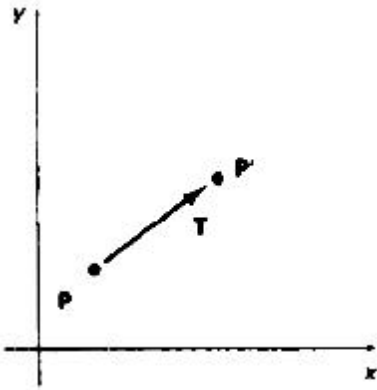


Fig. Translation of a point from p to p'

Translation is a rigid-body transformation that moves objects without deformation. That is, every point on the object is translated by the same amount. A straight Line segment is translated by applying the transformation equation to each of the line endpoints and redrawing the line between the new endpoint positions. Polygons are translated by adding the translation vector to the coordinate position of each vertex and regenerating the polygon using the new set of vertex coordinates and the current attribute settings.

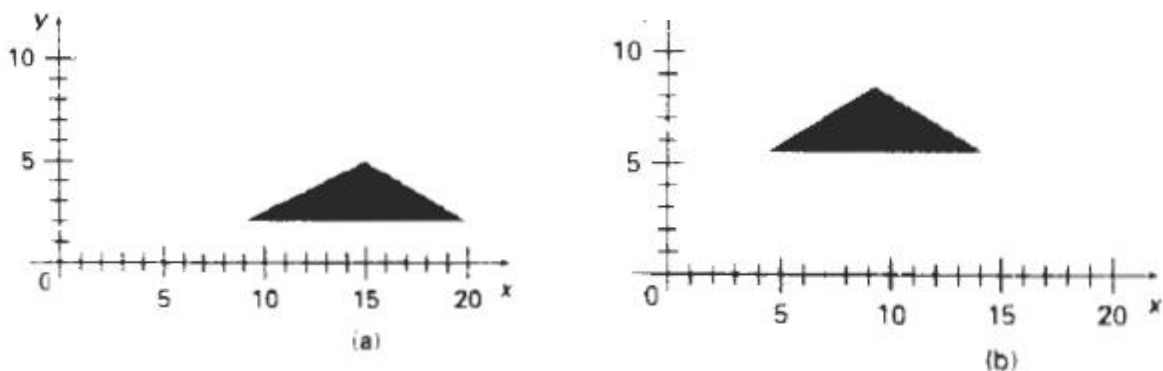


Fig.moving of an object from a to b.

Rotation

A two-dimensional rotation is applied to an object by repositioning it along a circular path in the xy plane. To generate a rotation, we specify a rotation angle θ and the position (x_r, y_r) of the rotation point (or pivot point) about which the object is to be rotated (Positive values for the rotation angle define counter clockwise rotations about the pivot point, as in Fig. , and

negative values rotate objects in the clockwise direction). This transformation can also be described as a rotation about a rotation axis that is perpendicular to the xy plane and passes through the pivot point.

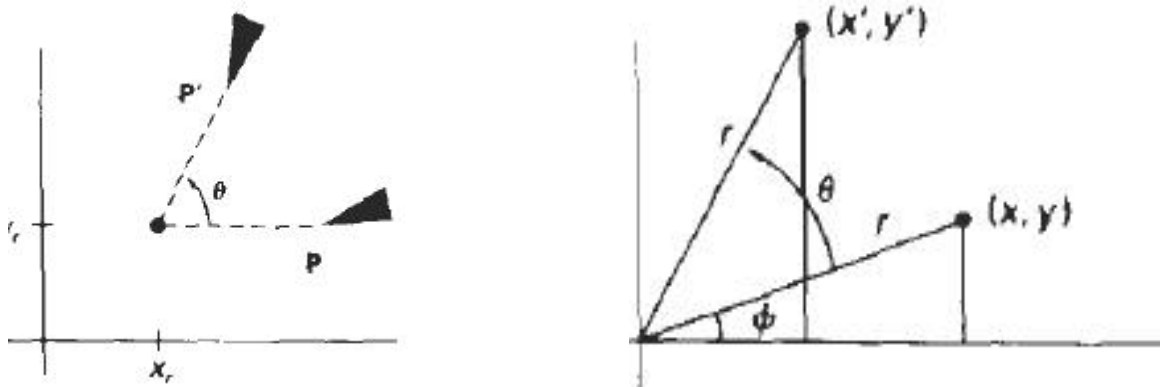


Fig .rotation of an object through angle θ .

$$x' = x \cos \theta - y \sin \theta$$

$$y' = x \sin \theta + y \cos \theta$$

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} \cos \Theta & -\sin \Theta & 0 \\ \sin \Theta & \cos \Theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

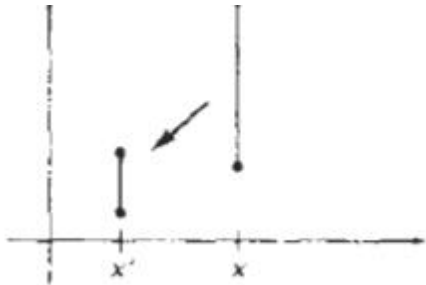
Scaling

A scaling transformation alters the size of an object. This operation can be carried out for polygons by multiplying the coordinate values (x, y) of each vertex by scaling factors s_x and s_y to produce the transformed coordinates (x', y') .



Fig. Turning a square(a) into rectangle (b)

Non-uniform scaling: different scalars per component:



Matrix form: S_x & S_y are scaling factors

Changes size.

$$x' = S_x \cdot x, \quad y' = S_y \cdot y$$

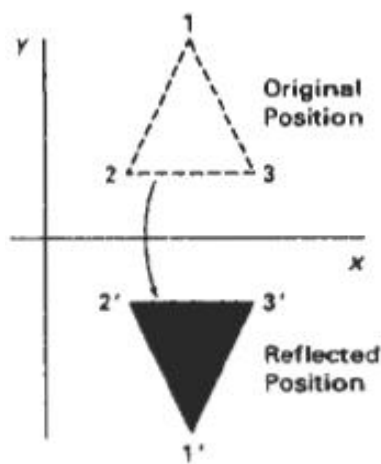
👉 Example:

- (2, 3), scale by 2
- (4, 6)

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} s_x & 0 & 0 \\ 0 & s_y & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Reflection

A reflection is a transformation that produces a mirror image of an object. The mirror image for a two-dimensional reflection is generated relative to an axis of reflection by rotating the object 180° about the reflection axis. We can choose an axis of reflection in the xy plane or perpendicular to the xy plane. When the reflection axis is a line in the xy plane, the rotation path about this axis is in a plane perpendicular to the xy plane. For reflection axes that are perpendicular to the xy plane, the rotation path is in the xy plane.



Reflection of an object about x-axis.

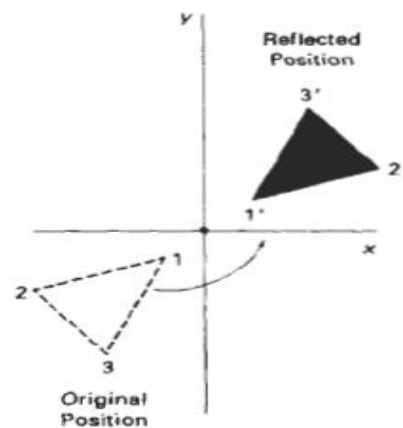
Mirror line: X-axis ($y = 0$)

Rule:

$$(x, y) \rightarrow (x, -y)$$

Matrix form:

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$



Reflection of an object
Relative an axis perpendicular
to xy plane .

Shear

A transformation that distorts the shape of an object such that the transformed shape appears as if the object were composed of internal layers that had been caused to slide over each other is called a shear. Two common shearing transformations are those that shift coordinate w values and those that shift y values.

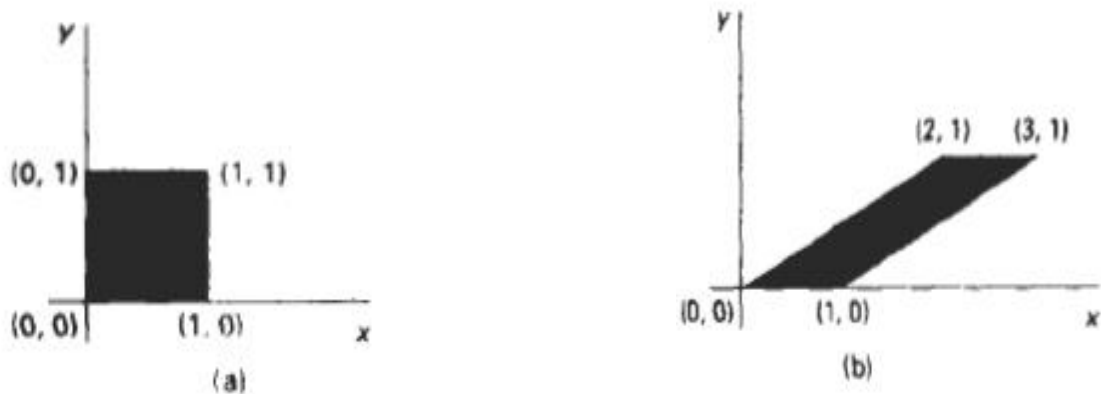


Fig.A unit square is converted into parallelogram

Three Dimensional Transformation

Methods for geometric transformations in three dimensions are extended from two-dimensional methods by including considerations for the z coordinate. We now translate an object by specifying a three-dimensional translation vector, which determines how much the object is to be moved in each of the three coordinate directions. Similarly, we scale an object with three coordinate scaling factors. The extension for three-dimensional rotation is less straightforward. When we discussed two-dimensional rotations in the xy plane, we needed to consider only rotations about axes that were perpendicular to the xy plane.

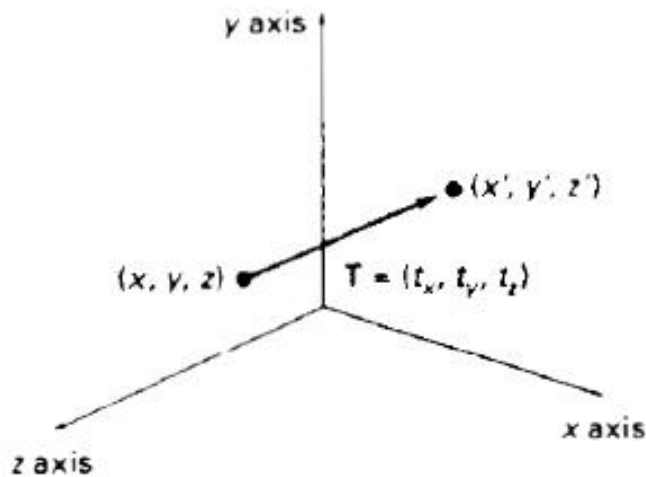
Translation

In a three-dimensional homogeneous coordinate representation, a point is translated from position $P = (x, y, z)$ to position $P' = (x', y', z')$ with the matrix operation

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & t_x \\ 0 & 1 & 0 & t_y \\ 0 & 0 & 1 & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

Parameters t_x , t_y and t_z specifying translation distances for the coordinate directions x' , y' and z' are assigned any real values. The matrix representation is equivalent to the three equations

$$x' = x + t_x \qquad y' = y + t_y \qquad z' = z + t_z$$



Translating a point with translation vector $T = (t_x, t_y, t_z)$

An object is translated in three dimensions by transforming each of the defining points of the object. For an object represented as a set of polygon surfaces, we translate each vertex of each surface and redraw the polygon facets in the new position.

Rotation

To generate a rotation transformation for an object, we must designate an axis of rotation about which the object is to be rotated and the amount of angular rotation. Unlike two-dimensional applications, where all

transformations are carried out in the xy plane, a three-dimensional rotation can be specified around any line in space. The easiest rotation axes to handle are those that are parallel to the coordinate axes. Also, we can use combinations of coordinate axis rotations to specify any general rotation.

Coordinate-Axis Rotations

1. Rotation Around X-axis

When rotating around the X-axis, the x-coordinate remains unchanged.

$$R_x(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{bmatrix}$$

Transformation:

$$x' = x$$

$$y' = y \cos \theta - z \sin \theta$$

$$z' = y \sin \theta + z \cos \theta$$

2. Rotation Around Y-axis

When rotating around the Y-axis, the y-coordinate remains unchanged.

$$R_y(\theta) = \begin{bmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{bmatrix}$$

Transformation:

$$x' = x \cos \theta + z \sin \theta$$

$$y' = y$$

$$z' = -x \sin \theta + z \cos \theta$$

3. Rotation Around Z-axis

When rotating around the Z-axis, the z-coordinate remains unchanged.

$$R_z(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Transformation:

$$x' = x \cos \theta - y \sin \theta$$

$$y' = x \sin \theta + y \cos \theta$$

$$z' = z$$

Example

Suppose point:

$$P(1, 1, 1)$$

Rotate it by 90° around the Z-axis.

Using:

$$x' = x \cos 90^\circ - y \sin 90^\circ$$

$$y' = x \sin 90^\circ + y \cos 90^\circ$$

Since:

$$\cos 90^\circ = 0, \quad \sin 90^\circ = 1$$

Result:

$$x' = (1)(0) - (1)(1) = -1$$

$$y' = (1)(1) + (1)(0) = 1$$

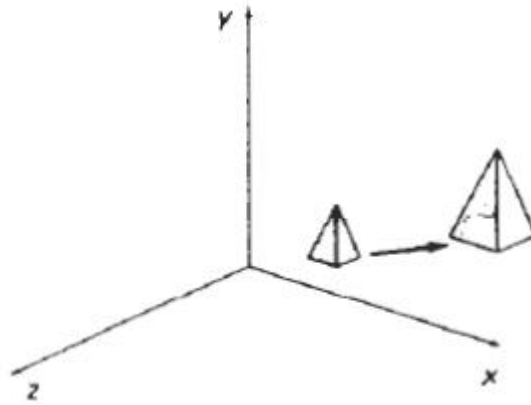
$$z' = 1$$

New point:

$$(-1, 1, 1)$$

Scaling

The matrix expression for the scaling transformation of a position $P = (x, y, z)$ relative to the coordinate origin can be written as



Doubling the size of an object with transformation also moves the object farther from the origin

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} s_x & 0 & 0 & 0 \\ 0 & s_y & 0 & 0 \\ 0 & 0 & s_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

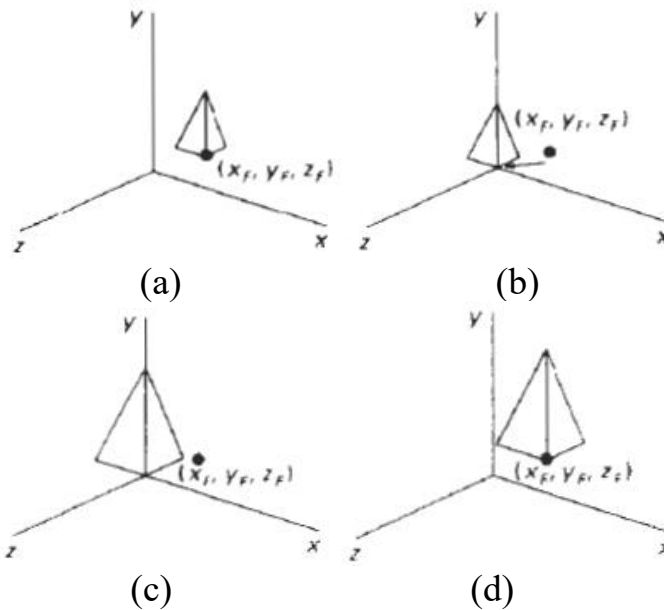
where scaling parameters s_x , s_y and s_z are assigned any positive values. Explicit expressions for the coordinate transformations for scaling relative to the origin are

$$x' = x \cdot s_x, \quad y' = y \cdot s_y, \quad z' = z \cdot s_z$$

Scaling an object with transformation changes the size of the object and repositions the object relative to the coordinate origin. Also, if the transformation parameters are not all equal, relative dimensions in the object are changed. We preserve the original shape of an object with a uniform scaling ($s_x = s_y = s_z$).

Scaling with respect to a selected fixed position (x_f , y_f , z_f) can be represented with the following transformation sequence:

1. Translate the fixed point to the origin.
2. Scale the object relative to the coordinate origin.
3. Translate the fixed point back to its original position.

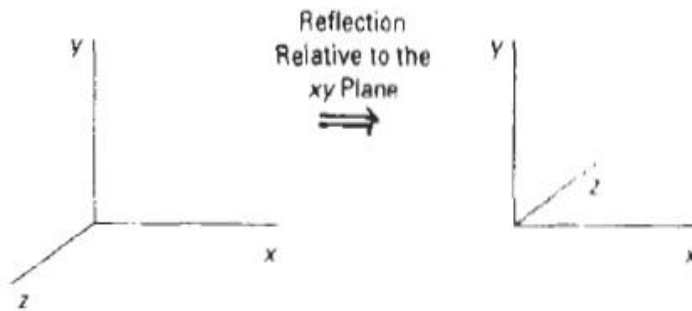


Scaling an object relative to a selected fixed point is equivalent to the sequence of transformations shown.

Reflection

A three-dimensional reflection can be performed relative to a selected reflection axis or with respect to a selected reflection plane. In general, three-dimensional reflection matrices are set up similarly to those for two dimensions. Reflections relative to a given axis are equivalent to 180° rotations about that axis. Reflections with respect to a plane are equivalent to 180° rotations in four-dimensional space. When the reflection plane is a coordinate plane (either xy, xz, or yz), we can think of the transformation as a conversion between Left-handed and right-handed systems.

An example of a reflection that converts coordinate specifications from a right-handed system to a left-handed system (or vice versa) is shown. This transformation changes the sign of the z coordinates, leaving the x and y-coordinate values unchanged. If xy plane then z will have opposite sign, if yz plane then x has opposite sign and if xz then y will have opposite sign. Transformation matrices for inverting x and y values are defined similarly, as reflections relative to the yz plane and xz plane, respectively. Reflections about other planes can be obtained as a combination of rotations and coordinate-plane reflections.



Conversion of coordinate specifications from a right handed to a left handed system can be carried out with the reflection transformation.

Shear

Shearing transformations can be used to modify object shapes. They are also useful in three-dimensional viewing for obtaining general projection transformations. In two dimensions, we discussed transformations relative to the x or y axes to produce distortions in the shapes of objects. In three dimensions, we can also generate shears relative to the z axis.

As an example of three-dimensional shearing, the following transformation produces a z-axis shear:



A unit cube (a) is sheared (b) by transformation matrix, with $a=b=1$.

Application

1. Used in animations formation
2. Science and Engineering field
3. Three dimensional display of object
4. Entertainment, advertising and graph presentation.